If Q is (c, d), then

$c = OQ' = OB' - Q'B'$

$= x_2 - \left(\frac{x_2 - x_1}{3}\right) = \frac{2x_2 + x_1}{3}$;

$d = OQ'' = OB'' - Q''B''$

$= y_2 - \left(\frac{y_2 - y_1}{3}\right) = \frac{2y_2 + y_1}{3}$

Thus the required point Q is $\left(\frac{2x_2 + x_1}{3}, \frac{2y_2 + y_1}{3}\right)$

Example 5.16 Find the points of trisection of the line segment joining $(-2, -1)$ and $(4, 8)$.

Solution Let $A(-2, -1)$ and $B(4, 8)$ are the given points.

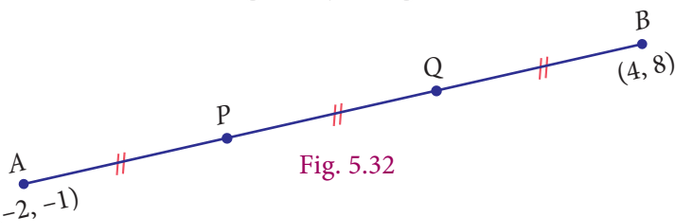


Fig. 5.32

Let $P(a, b)$ and $Q(c, d)$ be the points of trisection of AB, so that $AP = PQ = QB$.

By the formula proved above,

P is the point

$\left(\frac{x_2 + 2x_1}{3}, \frac{y_2 + 2y_1}{3}\right) = \left(\frac{4 + 2(-2)}{3}, \frac{8 + 2(-1)}{3}\right)$

$= \left(\frac{4 - 4}{3}, \frac{8 - 2}{3}\right) = (0, 2)$

Q is the point

$\left(\frac{2x_2 + x_1}{3}, \frac{2y_2 + y_1}{3}\right) = \left(\frac{2(4) - 2}{3}, \frac{2(8) - 1}{3}\right)$

$= \left(\frac{8 - 2}{3}, \frac{16 - 1}{3}\right) = (2, 5)$

Progress Check

(i) Find the coordinates of the points of trisection of the line segment joining $(4, -1)$ and $(-2, -3)$.

(ii) Find the coordinates of points of trisection of the line segment joining the point $(6, -9)$ and the origin.

5.6 Section Formula

We studied bisection and trisection of a given line segment. These are only particular cases of the general problem of dividing a line segment joining two points (x_1, y_1) and (x_2, y_2) in the ratio $m : n$.

Given a segment AB and a positive real number r.

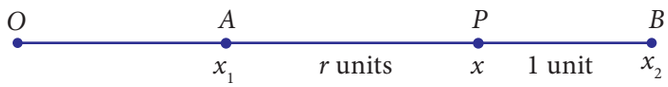


Fig.5.33

We wish to find the coordinate of point P which divides AB in the ratio r :1. This means $\frac{AP}{PB} = \frac{r}{1}$ or $AP = r(PB)$.

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